

Task 16: Debugging with GDB

1. Buggy C Program (task16.c)

```
#include <stdio.h>

int main() {

    int i = 0;

    int sum = 0;

    // Logical error: should be i < 10
    while (i > 10) {

        sum += i;

        i++;

    }

    printf("Sum is: %d\n", sum);

    return 0;

}
```

2. Compilation with Debug Symbols

Use the -g flag to include debugging information:

```
gcc -g task16.c -o task16
```

3. Running GDB

Task 16: Debugging with GDB

Run GDB with the compiled program:

```
gdb ./task16
```

Inside GDB, use the following commands:

```
(gdb) break main      # Set breakpoint at main
```

```
(gdb) run             # Start program
```

```
(gdb) step            # Step through lines
```

```
(gdb) print i         # Inspect variable i
```

```
(gdb) print sum       # Inspect variable sum
```

```
(gdb) list            # Show source code
```

4. The Bug

The loop condition `i > 10` is incorrect. It should be `i < 10`.

5. Fixing the Code

```
// Corrected loop condition
```

```
while (i < 10) {
```

```
    sum += i;
```

```
    i++;
```

```
}
```

6. Expected Output

Task 16: Debugging with GDB

After correction, recompile and run:

```
gcc -g task16.c -o task16
```

```
./ task16
```

Expected output:

```
Sum is: 45
```